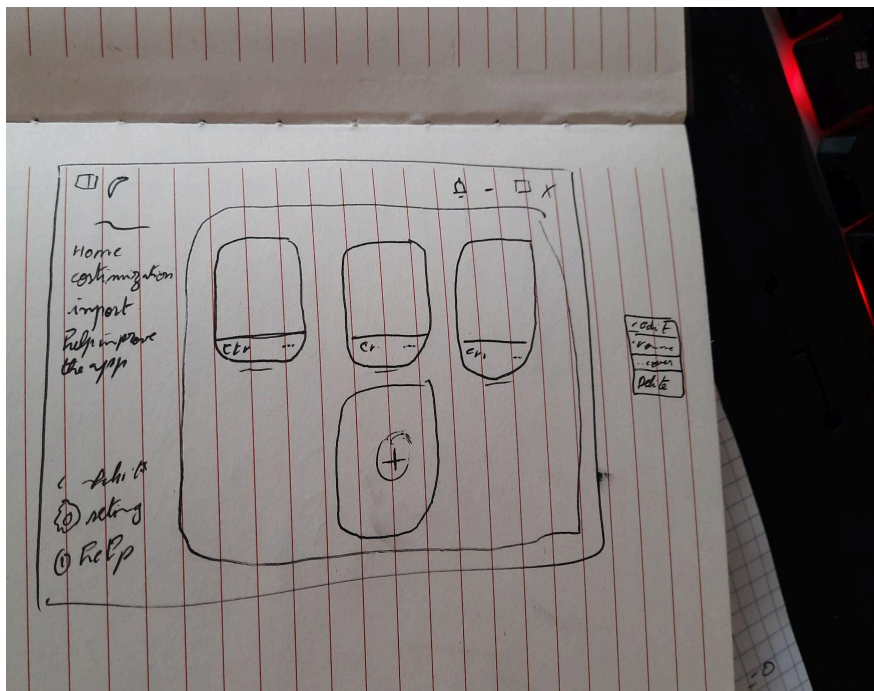


Biomes — Lightweight Desktop Workspace Manager

Biomes is an open-source, lightweight desktop utility that empowers users to create, save, and switch between customized visual workspaces tailored to specific workflows—such as coding, video editing, or writing.

Key Features

- **Custom Workspaces ("Biomes"):** Group your application layouts into dedicated environments. Easily switch between an "Editing Biome," a "Coding Biome," or a "Writing Biome" on the fly.
- **Visual Dashboard (Home Page):** Display all your saved Biomes as custom visual tiles.
 - Customize each Biome with a custom **Name**, **Cover Image**, and **Global Hotkey**.
 - Edit layout configurations, hotkeys, or artwork at any time from an intuitive interface.
- **Tiling & Grid Layout Engine:** Inspired by **Hyprrland**, Biomes lets you construct custom window layouts using two intuitive methods:
 - **Split Method:** Start with a full-screen canvas and interactively divide regions into smaller sub-containers.
 - **Grid Builder:** Draw or snap multi-region grid layouts directly on your display.
- **Fast & Lightweight:** Built to consume minimal system resources, keeping your system fast and your focused workflow distraction-free.



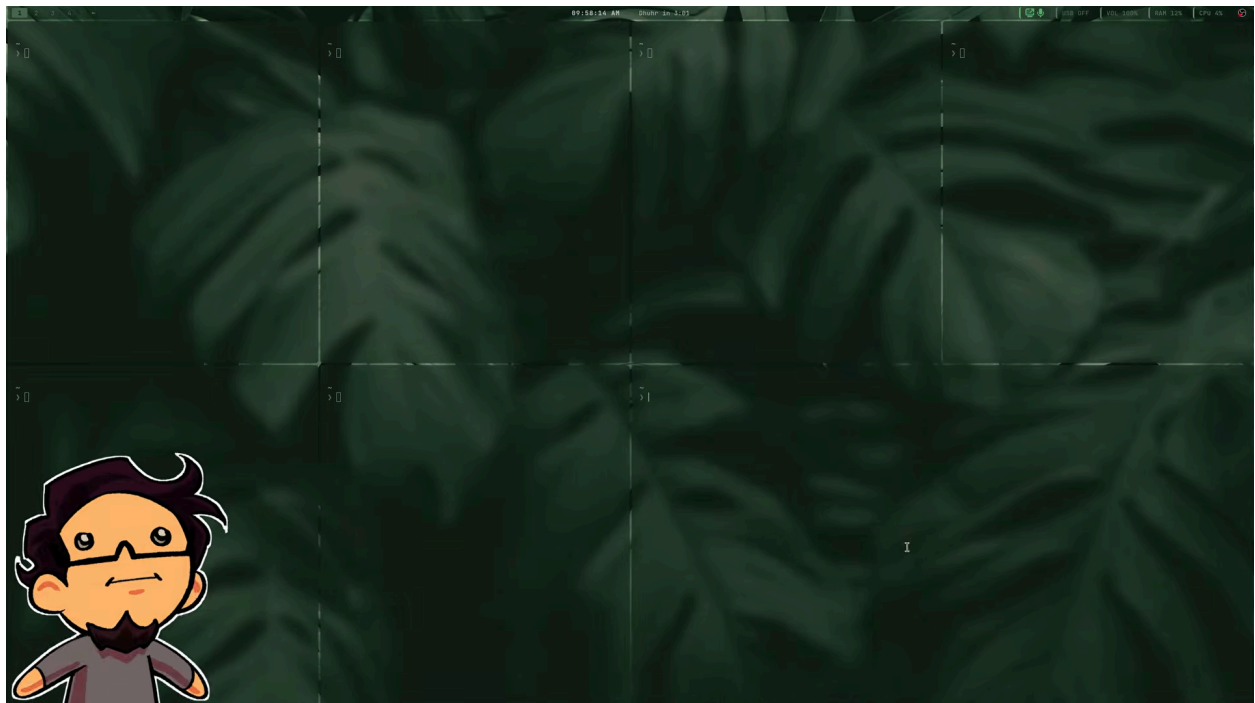
Method 1: Matrix Grid Overlay (WindowGrid Style)

1. **Grid Generation:** The user opens the creation interface and selects grid dimensions (e.g., 2×2 , 3×2 , or custom rows/columns with padding gaps).
2. **Visual Overlay:** The app renders a clean full-screen overlay (like in your screenshot) showing clear target boxes on top of their actual monitor wallpaper.
3. **Drag & Drop Assignment:** The user drags their open app windows (or picks executables) and drops them into specific boxes.
4. **Customization & Save:** The user gives the Biome a name (e.g., "Coding & Research"), assigns a custom global hotkey (e.g., CTRL+ALT+C), picks an icon/image, and saves it to `coding_biome.json`.

Method 2: Dynamic Split Canvas (Hyprrland Style)

1. **Interactive Splitting:** The user starts with a full-screen canvas on their monitor.
2. **Sub-Division:** Clicking on any box allows them to split that specific box horizontally (H) or vertically (V) into smaller sub-boxes.
3. **Drag & Drop Assignment:** Apps are dragged and dropped into these custom non-uniform boxes.
4. **Customization & Save:** The custom layout is saved with a name and global hotkey.

In the two plane the user end up with clear boxes on the screen looking like this



(in the image ignore the sticker in the left bottom)

And the user will drag and drop apps from the task bar or from the windows search

When the user completes building his biome he can open it or close it by the hotkey or the box in the home screen of the app. Note (when the user clicks on the add a new biome all the apps

go in the task bar like he click **Win + D**)

Visual Mapping & One-Key Switching

After splitting your screen into clear layout zones, simply drag and drop applications from your taskbar or Windows Search into your target containers.

Once saved, launch or close your entire workspace instantly using your custom **Hotkey** or dashboard tile. Launching a new Biome automatically minimizes background apps to keep your workspace distraction-free.

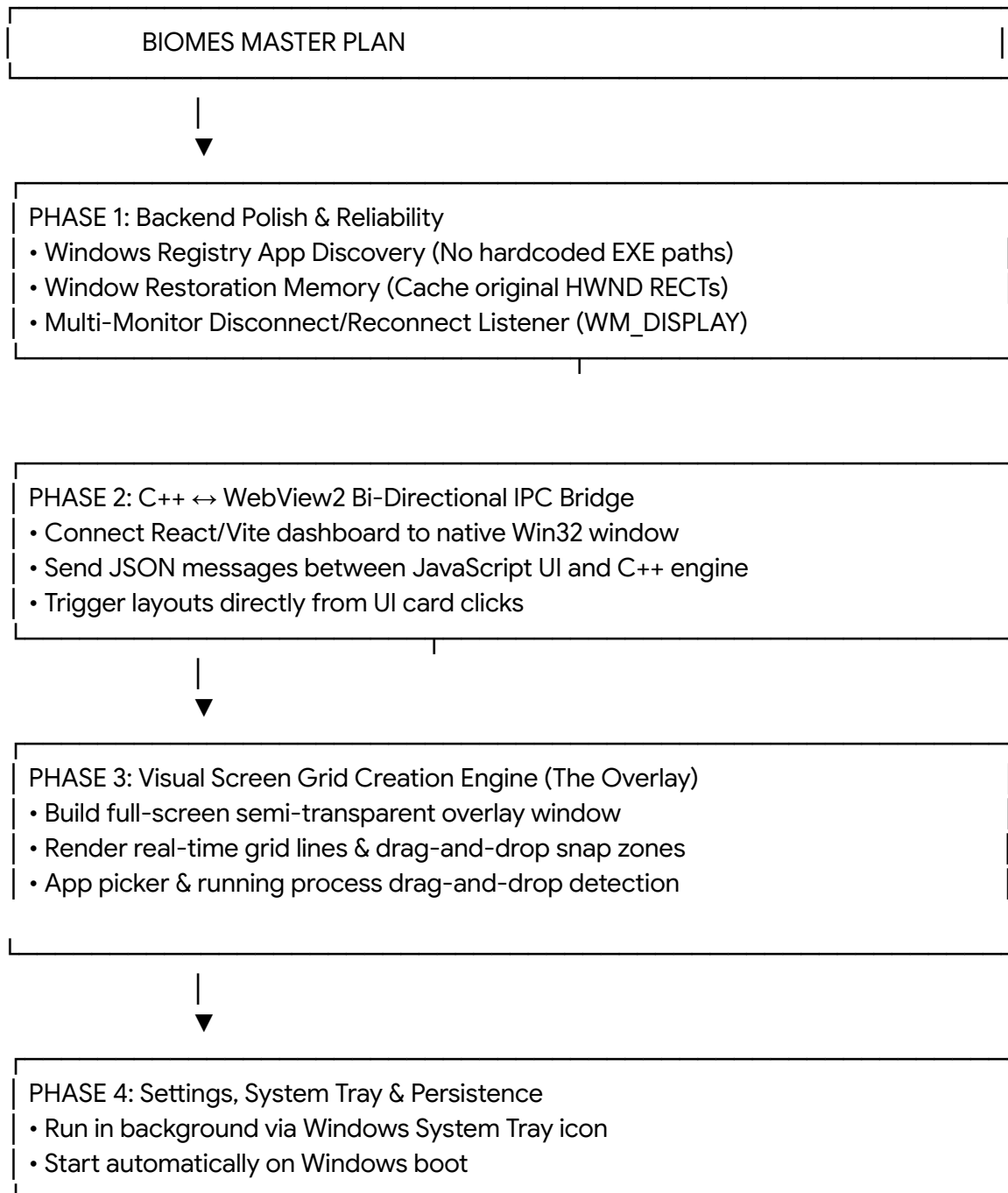
Futcher to make the app like a native windows app:

Multi-Display & Dynamic Resolution Handling

1. **Per-Display Workspace Scoping**
 - Displays are treated independently, while multi-screen configurations are saved under a single **Biome Layout**.
 - When opening a Biome on a single-screen setup (like a laptop on the go), only the primary display layout activates—preventing multi-monitor layouts from merging onto a single screen.
2. **Responsive Screen Scaling**
 - Workspace layouts automatically adapt to different screen sizes and resolutions (e.g., switching between an external desktop monitor and a laptop display).
 - Container boundaries scale proportionally to maintain the intended workflow layout regardless of display aspect ratio or resolution.
3. **Smart Display Disconnection & Fallbacks**
 - Unplugging a monitor while a Biome is active gracefully minimizes its associated applications to the taskbar.
 - If a multi-screen Biome is launched on a single-monitor setup, Biomes automatically falls back to rendering only the active primary screen layout.
4. **Non-Intrusive Normal Window Behavior**
 - Closing a Biome resets application window states back to their native system defaults.
 - Launching an application independently outside of an active Biome opens it in its standard windowed/maximized size, rather than forcing its saved Biome layout position.

Master Implementation Plan

Here is the step-by-step master plan to build the entire system from where we are now to a complete, published application:



Detailed Breakdown of Upcoming Steps

Phase 1: Backend Polish (Immediate Next Step)

Before building the visual drag-and-drop overlay, the C++ engine needs three core capabilities:

1. **Dynamic App Resolver:** When a user assigns Obsidian.exe or Chrome.exe, C++ queries HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths so users never have to manually specify full path strings (C:\Users\...).
2. **Window Restoration Cache:** C++ caches `g_OriginalWindowPositions[hwnd]` before snapping so that when a Biome is untoggled or closed, windows restore to their exact prior positions instead of staying stuck in small layout sizes.
3. **Monitor Change Listener:** Handle `WM_DISPLAYCHANGE` messages so plugged/unplugged monitors update active grid layouts instantly.

Phase 2: WebView2 IPC Bridge

1. Pass JSON payloads from C++ to the React frontend (e.g., sending active monitor details, list of running windows, and saved Biome cards).
2. Listen for JS events (e.g., user clicks "Apply Biome" or "Create New Biome") in C++ using `ICoreWebView2::add_WebMessageReceived`.

Phase 3: Drag-and-Drop Creation Overlay

1. Create a transparent Win32 overlay window (`WS_EX_LAYERED | WS_EX_TOPMOST`).
2. Implement **Method 1 (Matrix Grid Overlay)** and **Method 2 (Dynamic Tree Splitter)** inside the UI so users can visually split screen zones, drag app icons into boxes, assign hotkeys, and click **Save**.

Another description to biomes

Biomes is a lightweight, high-performance Windows workspace automation engine and dynamic window manager. It allows users to define, save, and instantly trigger screen layouts across multi-monitor setups.

The plane that we are following now 😊

PDF Requirement / Feature	Mechanism & Implementation Plan	Roadmap Phase
1. Restoration Cache (Non-Biome state)	C++ caches <code>g_OriginalWindowPositions[hwnd]</code> before applying any Biome layout. When untoggled/closed, <code>RestoreAllCapturedWindows()</code> puts windows back to their exact prior pixel coordinates.	Phase 1 & 2 (Already Built & Verified)

memory)		
2. Dynamic App Resolver	<code>WindowScaler::ResolveAppPath(processName)</code> searches <code>HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths</code> so users only need to specify <code>Obsidian.exe</code> or <code>chrome.exe</code> without full disk paths.	Phase 1 <i>(Already Built)</i>
3. Method 1: Matrix Grid Overlay (WindowGrid style)	Split display into uniform columns/rows (2×2 , 3×3 , etc.). Drag across cells to merge region coordinates into a target <code>SelectedBox</code> .	Phase 3a <i>(Up Next)</i>
4. Method 2: Dynamic Tree Splitter (Hyprrland / i3 style)	Binary space partitioning (BSP tree). Interactively split boxes horizontally or vertically (<code>\text{Parent} \rightarrow \text{Left/Right}</code> or <code>\text{Top/Bottom}</code>).	Phase 3b
5. Drag-and-Drop Container Assignment	HTML5 Drag & Drop interface where app tiles (fetched via <code>GET_ACTIVE_WINDOWS</code>) are dropped directly onto grid/split zones in the visual editor.	Phase 3c
6. Biome Persistence & JSON Storage	<code>JsonManager</code> serializes created workspace profiles to disk (<code>config/biomes.json</code>) with metadata: Name, Cover Image, Hotkey, display bounds, and app placement mapping.	Phase 4a

7. Multi-Monitor & Responsive Layout Listener	MonitorManager handles WM_DISPLAYCHANGE to scale percentage-based grid coordinates dynamically across different monitor DPIs and resolutions.	Phase 4b
8. Global Hotkeys	HotkeyManager uses Win32 RegisterHotKey() to trigger Biomes instantly with keyboard shortcuts (e.g., Ctrl+Alt+C for Coding Biome).	Phase 4c